Mr Dimelow

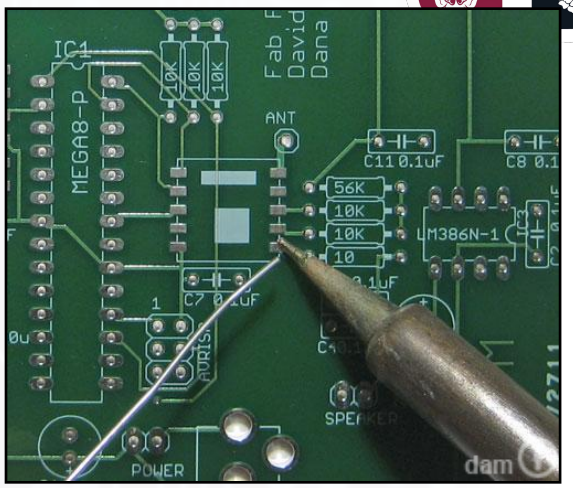
3 Processes, techniques and specialist tools

3.4 Permanent and Semi-Permanent Joining Techniques

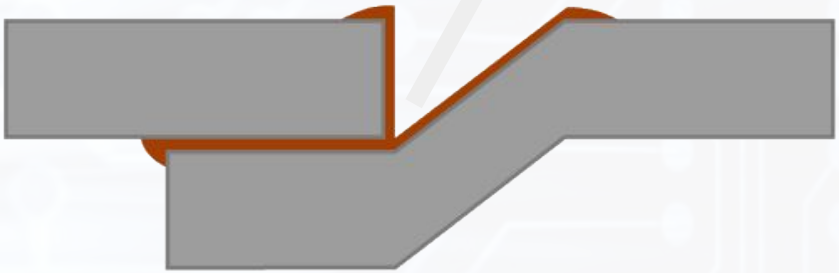
c) heat – oxy-acetylene welding, MIG welding, brazing, hard soldering, soft soldering (including use of specialist tools)

Joining with Soldering





Heat



In this presentation you will:

- see how metals can be joined by brazing or soldering

Next >

Introduction

Brazing and soldering are both ways of joining metals using a molten filler metal.

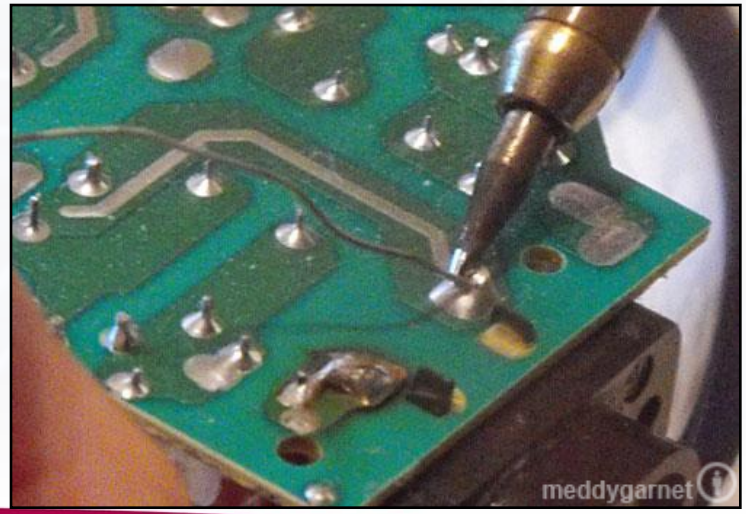
The filler bonds with the surfaces, cools and solidifies.



Brazing

Unlike welding, the pieces being joined ***do not*** melt.

Brazing uses higher temperatures than soldering.



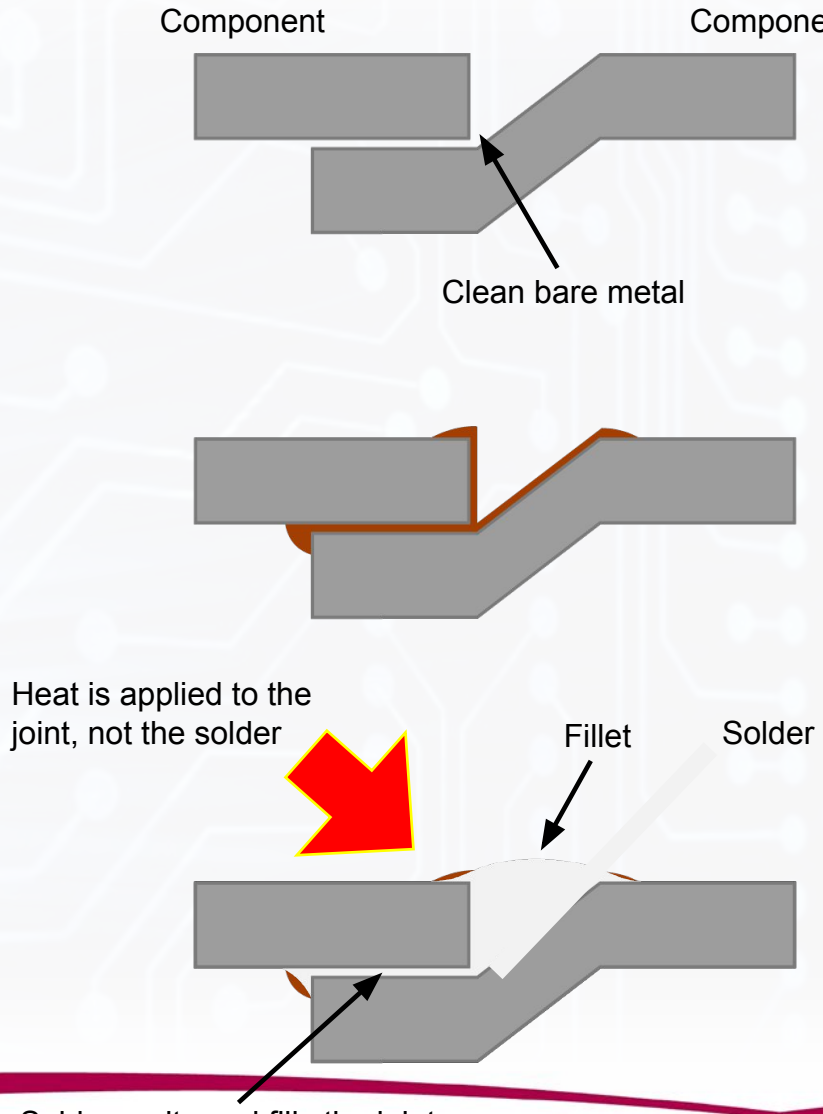
The Brazing or Soldering Process

The joining surfaces must be clean.

Flux is applied to prevent oxidation of the joining surfaces.

Heat and solder are applied. Molten solder fills the joint.

The joint cools and the solder bonds with the surfaces.

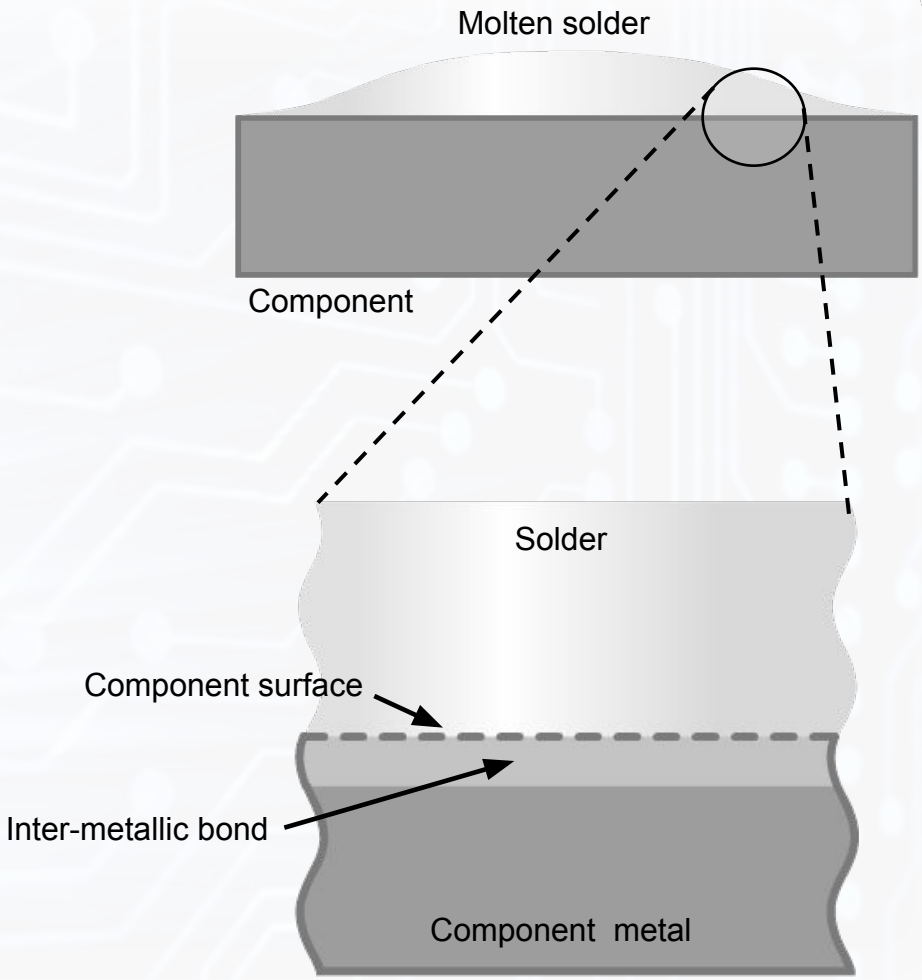


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Solder Wetting and Bonding

When solder wets the component surface, a new alloy metal is created within the surface material.

A strong bond forms between the solder and the component metal.

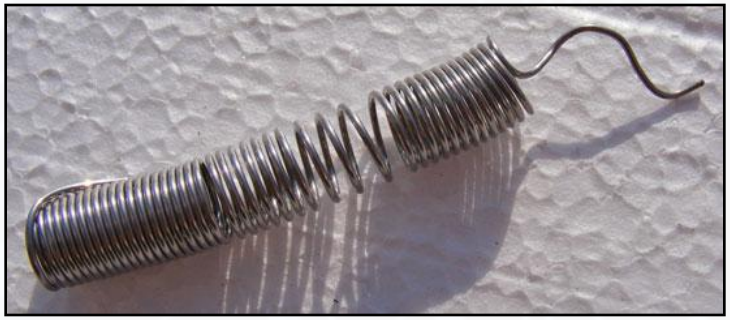


Solders

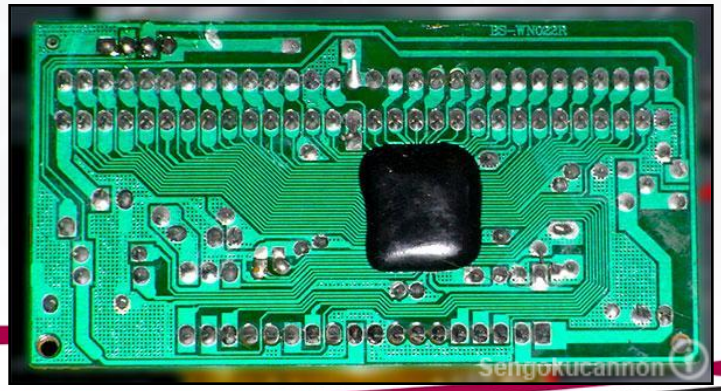
Solders are alloys designed to have a lower melting point than the metals to which they bond.



Some solders are eutectic alloys that melt at a specific temperature with no semi-molten state.



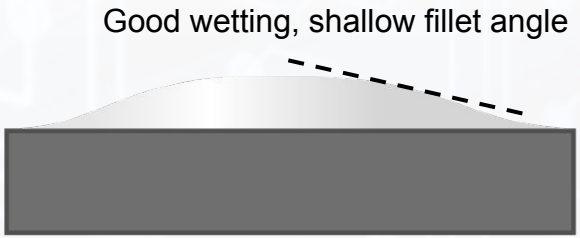
This is useful for electronics assembly techniques.



Next >

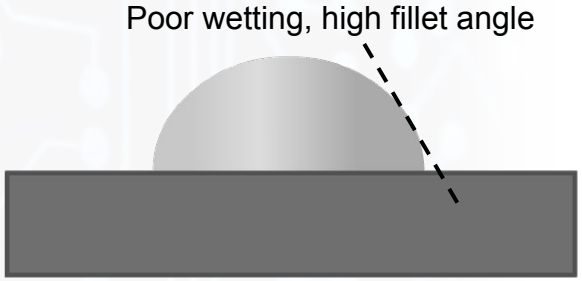
Cleaning and Oxide Removal

Joint surfaces should be clean and free from oxides.



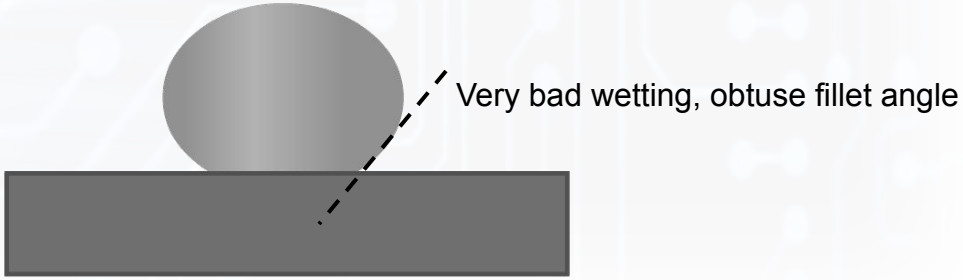
Clean and oxide free

This allows wetting of the joint surfaces, good solder flow and shallow fillet angles.



Dirt and/or light oxidisation

Dirt and oxidation cause high fillet angles and a poor, ‘dry’ joint.



Very dirty and/or heavy oxidisation

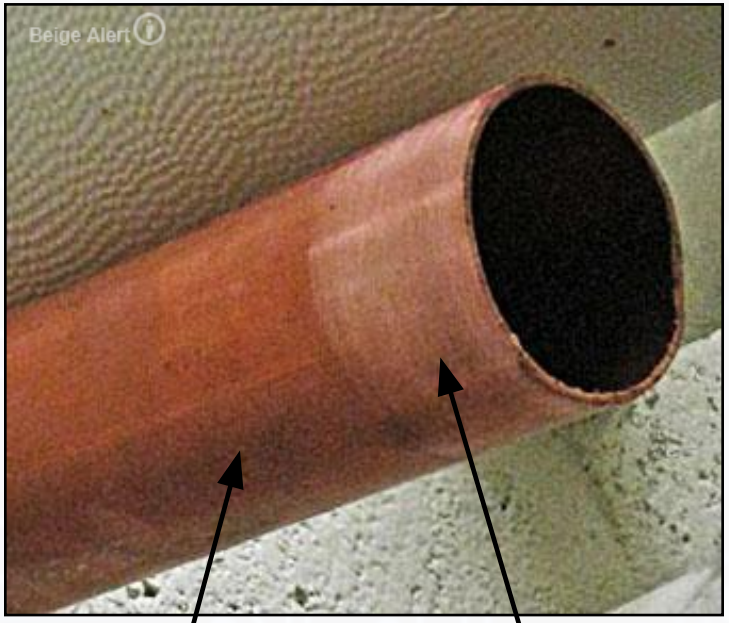
Next >

Cleaning and Oxide Removal

Handling metals leaves dirt and oily deposits on their surfaces.

Most metals react with oxygen in the air to form oxides on their surfaces.

The dirt and oxides must be removed before brazing or soldering.



Discolouration caused by oxidisation and dirt

Cleaning with emery cloth reveals clean bright material

Next >

Applying Flux

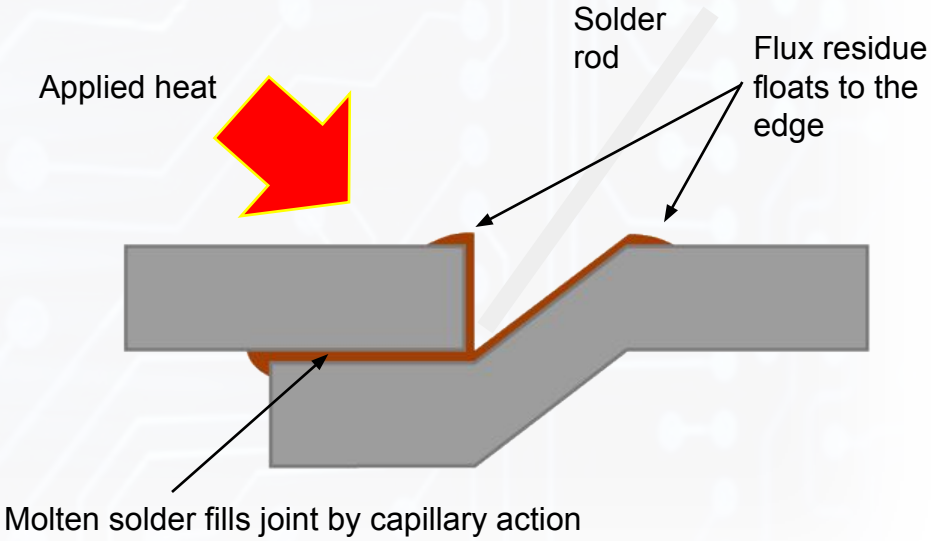
Flux melts at a lower temperature than solder.



Flux paste

It forms a protective layer that prevents applied heat causing further oxidation.

When molten solder fills the joint, the flux floats away and forms a residue around the edge.



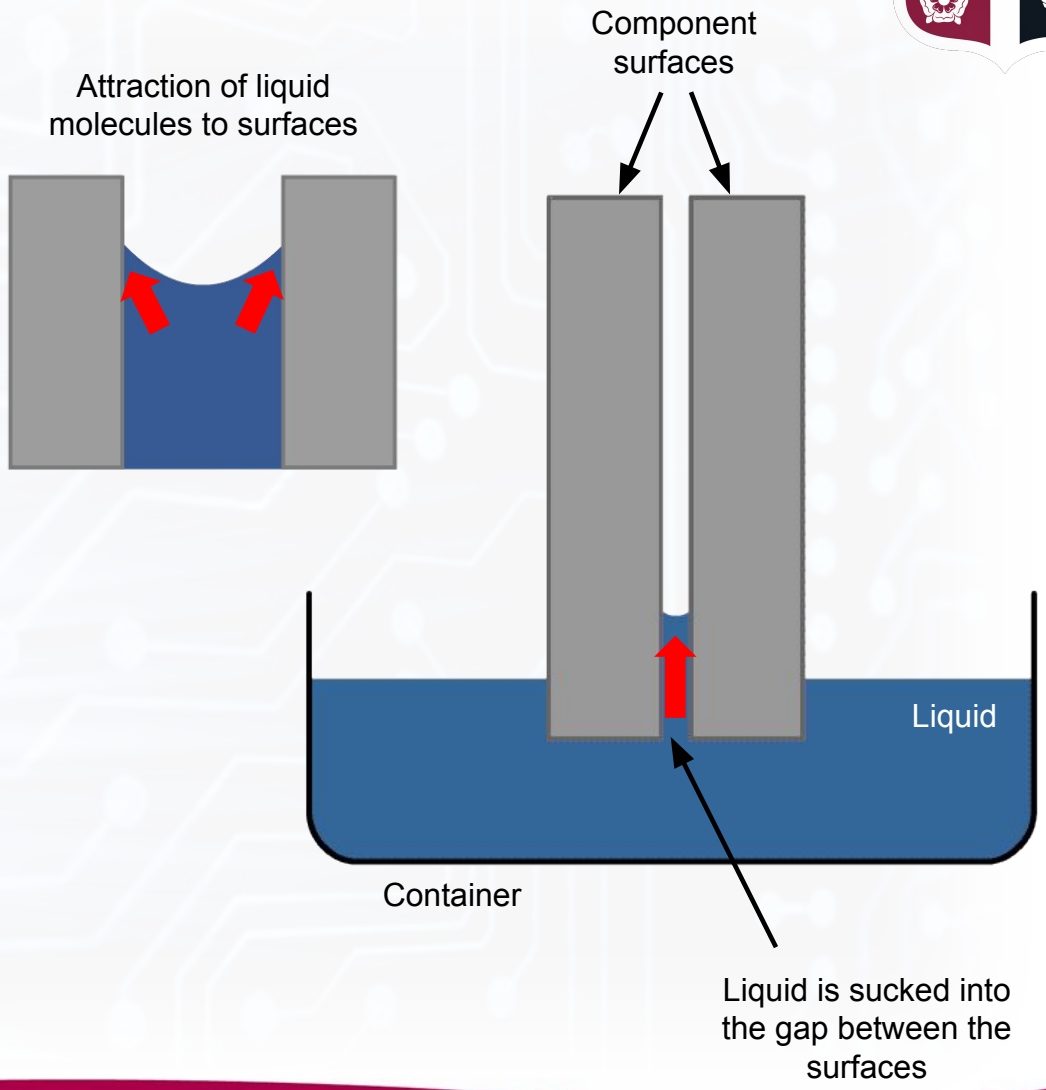
Next >

Capillary Action

Liquid flows naturally into a narrow gap, even against gravity. This is called capillary action.

It is caused by inter-molecular attraction between liquid and solid surfaces.

It causes solder to flow into a joint.



Next >

Heat Sources

Brazing uses higher temperatures than soldering.

Oxy-gas torches are used for brazing in metal fabrication and plumbing.

Portable propane blowtorches are used for plumbing.

Electric soldering irons are used for small metalworking and electronics assembly.

700°C

250°C



Oxy-acetylene brazing torch



Hand-held gas blowtorch



Heavy-duty electric soldering iron



Small electric soldering iron

Next >

Metals that can be Brazed or Soldered

Some metals that are unsuitable for welding can be brazed instead.

Unlike welding, brazing can create joins between different metals.

Brazing is less expensive than welding but not as strong.



Copper



Brass



Aluminium



Titanium

Next >

Brazing Bike Frames

Brompton Bicycles



Question 1

“In brazing and soldering, the metal surfaces to be joined are melted together.”

Is this statement true or false?



Next >

Question 1

“In brazing and soldering, the metal surfaces to be joined are melted together.”

Is this statement true or false?



False

The metal surfaces do not melt. They are bonded by a molten metal alloy (solder).

Next >

Question 2

What is the purpose of adding flux during the brazing or soldering process?

A) To keep the temperature low

B) To keep the temperature high

C) To clean and protect the joint surfaces

D) To clean the joint when it has cooled



Next >

Question 2

What is the purpose of adding flux during the brazing or soldering process?

A) To keep the temperature low

B) To keep the temperature high

C) To clean and protect the joint surfaces

D) To clean the joint when it has cooled



Flux is applied to protect the joint from oxidation during heating.

Next >

Question 3

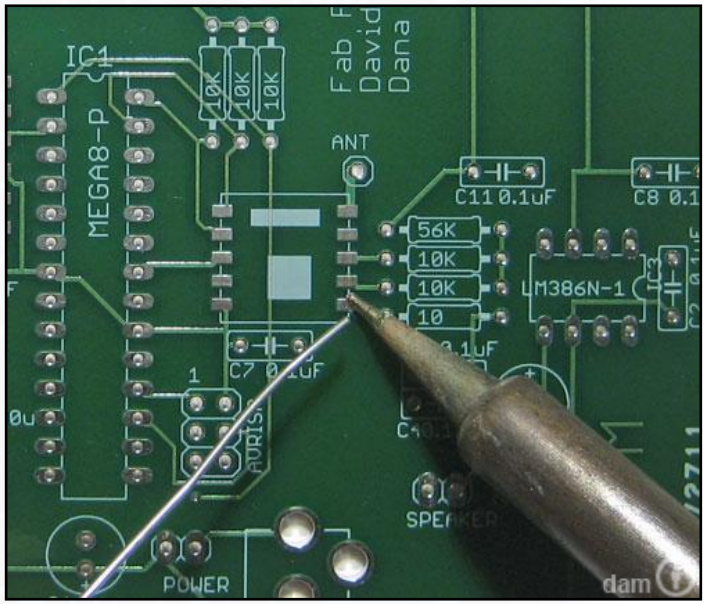
When does solder show good wetting on a metal surface?

A) When it solidifies quickly as it cools

B) When it melts easily at low temperatures

C) When it stays in one place and forms a ball shape

D) When it flows and bonds with the surface easily



Question 3

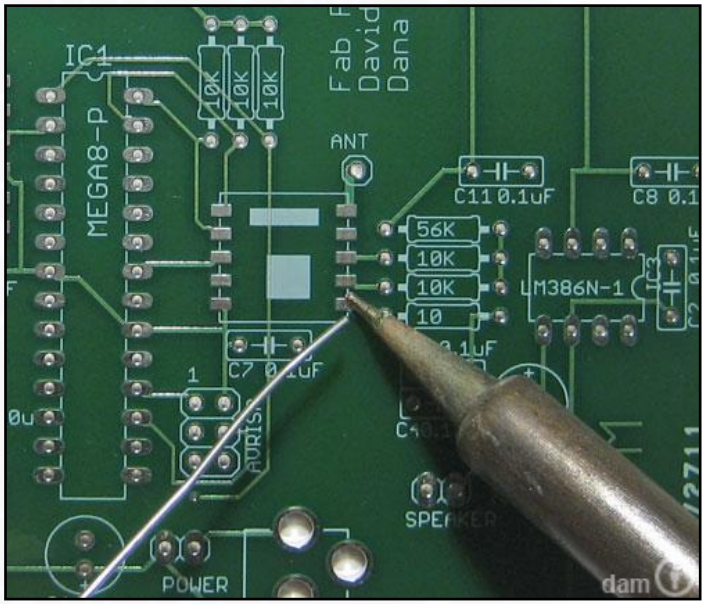
When does solder show good wetting on a metal surface?

A) When it solidifies quickly as it cools

B) When it melts easily at low temperatures

C) When it stays in one place and forms a ball shape

D) When it flows and bonds with the surface easily



Solder will only wet a metal surface if it flows and bonds with it easily.

Next >

Summary

In this presentation you have seen:

- how metals can be joined by brazing or soldering